

The Poisson Distribution

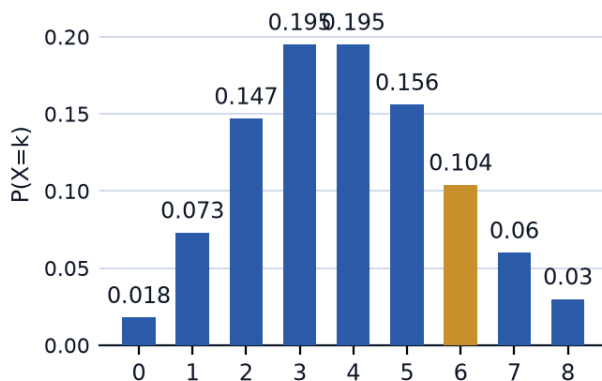


Conditions for a Poisson distribution

- 1 State the key conditions required for a random variable to be modeled by a Poisson distribution.
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Computing Poisson probabilities

- 2 A call center receives an average of 4 calls per minute, modeled by a Poisson distribution with $\lambda = 4$. Find $P(X = 6)$, using $P(X = k) = \frac{e^{-\lambda} \lambda^k}{k!}$, to three decimal places.



- 3 Using the same call center ($\lambda = 4$ calls per minute), find the probability of receiving at most 2 calls in a given minute.
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Mean and variance of a Poisson distribution

- 4 State the mean and variance of a Poisson distribution with parameter λ , and explain what is distinctive about this relationship compared to other common distributions.
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Sums of independent Poisson variables

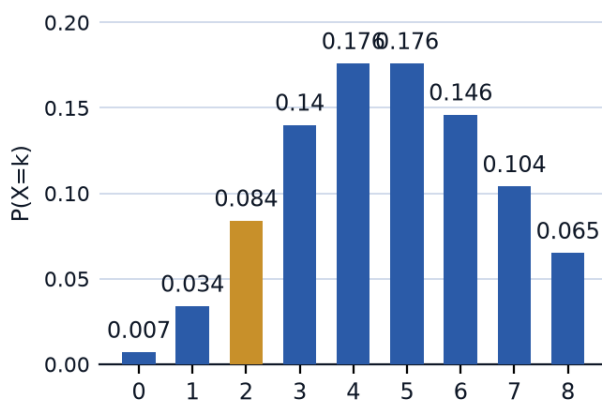
- 5 Two independent Poisson processes have rates $\lambda_1 = 3$ and $\lambda_2 = 5$ events per hour. If $X_1 \sim \text{Po}(3)$ and $X_2 \sim \text{Po}(5)$, state the distribution of $X_1 + X_2$, and find its mean and variance.
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Poisson as an approximation to the binomial

- 6 Explain the conditions under which a binomial distribution $X \sim B(n, p)$ can be well-approximated by a Poisson distribution, and state the appropriate value of λ to use.
- 7 A factory produces 2000 items, each independently defective with probability 0.002. Using the Poisson approximation, find the probability that exactly 3 items are defective, to three decimal places.

Using cumulative Poisson probabilities

- 8 A shop receives customers at an average rate of $\lambda = 5$ per 10-minute interval. Find the probability of receiving more than 2 customers in a given 10-minute interval, given $P(X = 0) \approx 0.0067$, $P(X = 1) \approx 0.0337$, and $P(X = 2) \approx 0.0842$.



Finding an unknown parameter from a given probability

- 9 A random variable $X \sim \text{Po}(\lambda)$ satisfies $P(X = 0) = 0.1353$. Find λ , to the nearest whole number, using $P(X = 0) = e^{-\lambda}$.

Variance-based reasoning with the Poisson distribution

- 10 A random variable $X \sim \text{Po}(\lambda)$ has standard deviation 3. Find λ , and then find $P(X = \lambda)$, to three decimal places.

Combining Poisson reasoning with real-world decisions

- 11 A website server crashes on average $\lambda = 0.5$ times per week. The IT team wants to know the probability of at least one crash in a given week, given $P(X = 0) \approx 0.6065$. Should the team be concerned if their target is to keep this probability below 30%?

Combining a Poisson approximation with conditional probability

- 12 Machine faults occur at an average rate of $\lambda = 1.5$ per day. Find the probability that exactly one fault occurs today, given that at least one fault occurs today, to three decimal places, using $P(X = 0) \approx 0.2231$ and $P(X = 1) \approx 0.3347$.

Combining Poisson processes over different time intervals

- 13 A radioactive source emits particles at a rate of $\lambda = 2$ per second, modeled by a Poisson distribution. Find $P(X = 3)$ for a 3-second interval, to three decimal places, using the fact that the parameter scales with the length of the interval.
- 14 Explain why the Poisson distribution is not an appropriate model for the number of heads obtained when flipping a fair coin 10 times, referencing the conditions required for a Poisson model.

Comparing two independent Poisson processes

- 15 Two independent Poisson processes have rates $\lambda_A = 3$ and $\lambda_B = 2$ per hour. Find the probability that, in a given hour, process A produces exactly 2 events and process B produces exactly 1 event, to three decimal places, given $P(X_A = 2) \approx 0.2240$ and $P(X_B = 1) \approx 0.2707$.

Rescaling the Poisson parameter for a different interval

- 16 A radio station receives an average of 12 requests per hour. Find the appropriate Poisson parameter λ to use for a 15-minute segment.

The mode of a Poisson distribution

- 17 A random variable $X \sim \text{Po}(4)$ has an integer parameter $\lambda = 4$. State the mode(s) of X (the most likely value or values), and explain why a Poisson distribution with an integer λ has two modes, using the ratio $P(X = k)/P(X = k - 1) = \lambda/k$.

Identifying an appropriate Poisson scenario

- 18 Which of the following scenarios is best modeled by a Poisson distribution?
- a. The number of typing errors on a randomly selected page of a manuscript.
 - b. The number of heads in 20 coin tosses.
 - c. The height of a randomly selected adult.
 - d. The number of correct answers on a 10-question multiple-choice test, each with 4 options.
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Assessing Poisson plausibility from sample data

- 19 A biologist counts the number of bacteria colonies in 10 equal-sized petri dishes: 3, 5, 4, 6, 5, 4, 3, 5, 6, 4. Find the sample mean and the unbiased sample variance, and comment on whether a Poisson model seems plausible for this data.